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# Theta, Alpha and Beta Activity in the Occipital Based on EEG Signals for Mental Fatigue in High School Students

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***Abstract*— Excessive physical and cognitive activity is believed to be the cause of a person experiencing mental fatigue. Meanwhile, one of the factors that cause acute illness is mental fatigue. Thus mental fatigue has been identified as one of the life-threatening factors in the human population, especially in high school students. In this paper, we use the EEG signal to analyze mental fatigue conditions in the occipital region. We also develop cognitive tasks as stimuli for mental fatigue. Theta, alpha, and beta waves activity from EEG signals are used to compare mental fatigue conditions with normal conditions in each subject. Afterward, the time domain is analyzed using the standard deviation and the mean absolute value (MAV). Meanwhile, the IOWA Fatigue Scale (IFS) questionnaire was used as a subjective reference. Finally, it was found that when the subject suffered mental fatigue an increase in the 2 features of the EEG signal was greater than normal conditions, between the pre-task and post-task in the occipital O1 and O2 channels. In general, an increase in the 2 features of the EEG signal occurs in theta, alpha, and beta waves. From this study by analyzing brain waves in the occipital area, it was concluded that high school students who attend school for approximately 8 hours a day have the potential to experience mental fatigue.**

***Keywords—Mental Fatigue, EEG Signals, Occipital, Theta waves, Alpha waves, Beta waves, IOWA Fatigue Scale, Mean Absolute Value, Standard Deviation.***

## I. Introduction

Fatigue, like sleepiness, is a very common phenomenon and a common symptom of various diseases, even fatigue can be observed in healthy individuals [1]. Fatigue is divided into two kinds, namely physical fatigue and mental fatigue. Physical fatigue is the result of repetitive muscular actions that cause muscle strength to be disrupted or weakened [1][2]. Meanwhile, mental fatigue can be interpreted as a failure to complete mental tasks that require self-motivation. Therefore, mental fatigue is associated with psychological responses that are indicated by feelings of excessive fatigue and difficult to continue to do the task [2].

Mental fatigue is now common in human populations. Those who suffer from mental fatigue are usually students and workers. Mental fatigue is now often associated with acute to chronic diseases that threaten human life [3]. In contrast to physical fatigue, mental fatigue is usually associated with feelings of excessive tiredness, increased anxiety, and also impaired cognitive function [2][4][5].

Currently, mental fatigue can be detected from two different techniques, namely psychological or physiological measurement techniques [5] where psychological measurement techniques are based on individual responses to the questionnaire given and physiological techniques based on measuring brain wave activity using EEG [5]. EEG signals are considered capable of detecting mental fatigue because mental fatigue is associated with significant changes in brain wave activity [6].

Another study mentioned, the evaluation of fatigue conditions through BCI was evaluated with flashing stimulus [4]. They put the EEG electrodes in the Oz channel position as the input channel [4]. But the method is only concentrated in the Oz position. Afterward in this study, we improved with the EEG Emotiv EPOC+ device that can detect brain signals in the occipital region by placing electrodes in O1 and O2 positions. The EEG Emotiv EPOC+ was considered capable of detecting fatigue conditions [7][8]. Moreover, they only use one stimulus flashing on a computer screen to induce fatigue [4]. Hence, we improved by designing new games. The new game is a cognitive task consisting of arithmetic tasks, the task of summarizing articles, and maze problem. Arithmetic task [9][10][11] and maze problem [5] stimuli have been used by some researchers to induce mental fatigue conditions. Based on several studies [9][12], they used a stimulus that relied on the subject's memory for something displayed on a computer screen. Therefore we adapted this by developing a new task model, the task of summarizing an article.

The Chalder Fatigue Scale (CFS) is used in previous studies to measure the physiological subject [4]. Because CFS only has 4 choices and 4 point scales. In this study, IOWA Fatigue Scale (IFS) was chosen as a physiological measurement for the subjects and as a validation of the subject's mental fatigue conditions. As has been proven by previous studies, IFS is considered able to show mental fatigue [5]. Because IFS has 5 choices and 5 point scales. Besides that IFS can classify the subject's condition into 3 levels, i.e. not fatigue, fatigue, and severe fatigue.

The previous investigation, delta (δ) wave were not believed to be able to show increased levels of fatigue, because although drowsiness and decreased arousal level can increase slow-wave brain activity, it turns out delta (δ) wave are very easily influenced by artifacts especially when the subject is in an awake condition [4]. Hence in this study, the delta (δ) waves are not used in the analysis of subject mental fatigue.

While theta (θ) waves are considered capable of showing the subject's mental fatigue [4][10][13][12]. Because when the subject feels tired, cognitive performance will decrease and followed by an increase in theta value [4]. Theta (θ) waves lie within the range of 4 – 8 Hz [14]. The very lowest waves of theta represent the fine line between being awake or

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in a sleep state [15]. Besides being related to cognitive efforts, theta rhythm is also related to working memory [11].

For alpha (α) wave will appear when the subject feels relaxed and sleepy but awake state [4]. Increased alpha (α) wave activity indicates increased efforts to maintain alertness levels [4][16]. Alpha (α) wave is also an indicator for cognitive and memory performance [17]. Alpha (α) waves lie within the range of 8 – 12 Hz [14]. Alpha waves are the basic rhythm of EEG signals [18]. Alpha waves are faster and shorter and are related to relaxation and disengagement [15]. Alpha waves occur when awake, especially in the occipital area of the brain, usually appearing while the eyes are closed and decreasing when the eyes are open and also weakening during attention to assigned tasks [19]. Previous research has shown that the strength of the alpha wave increases significantly between the beginning and end of the assigned task [11].

Beta (β) waves are associated with alertness, arousal, and excitement [4]. The increase in beta (β) wave activity indicates an attempt to a high level of alertness and arousal [6][13][16]. In several previous studies, the activity of theta (θ), alpha (α), and beta (β) waves that indicate a state of mental fatigue occurs in the occipital region [4][13][16]. Beta (β) waves lie within the range of 12 – 30 Hz. Beta waves are said to be small and fast, related to focused concentration, alertness and mental activity [15][18]. Moreover, beta waves appear when completing mathematical tasks [15].

Previous studies mentioned an increase in beta waves (β) from normal conditions using the Pearson correlation coefficient on the O1 and O2 channels [2]. Other studies suggest that alpha (α) and theta (θ) waves increase rapidly and reach maximum amplitude after working on cognitive tasks [12]. And also other studies have shown an increase in theta (θ), alpha (α), and beta (β) wave activity in all posterior areas or the positions of O1 and O2 electrodes [6]. But problems arise in beta and alpha waves, where one study states that compared to normal conditions, the mean value of Approximate Entropy (ApEn) and Kolmogorov complexity (Kc) in all electrodes decreased significantly ($P<0.05$) in beta waves (β) and alpha (α), while theta (θ) and delta waves do not change significantly [18].

In this study, we propose objective measurements of theta, alpha, and beta waves in the O1 and O2 regions with an approach based on EEG signals and subjectively with validation from IFS. The subject research focused on high school students who went through the learning process in school for approximately 8 hours a day. It is hoped that this research can prove the mental fatigue conditions experienced by students. Meanwhile, in processing information from recorded EEG signals, we use the time-domain approach. Where the time-domain approach is considered appropriate for analyzing and comparing subject conditions with measurement time. At last, with the use of new cognitive tasks such as the task of summarizing articles and maze problem, as well as the placement of electrode positions which are considered to be more ideal using O1 and O2 channels, it is expected to know with certainty the activity of the brain waves.

## II. Material and Method

### *2.1 Material*

In this study, we used wireless EEG from EMOTIV EPOC+. This EMOTIV EPOC+ transmits data at 256 Hz without being tethered to a computer. And also EMOTIV EPOC+ has 14 channels namely AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8 and AF4 with references in P3 / P4. Also, EMOTIV EPOC+ has a bandwidth of 0.16 - 43 Hz, with a digital notch filter at 50 Hz and 60 Hz.

### *2.2 Method*

#### *A. Experimental Subjects*

The subjects in this study came from high school students with a learning system in school for 8 hours, start from 06.45 am and ends at 03.00 pm. Thirteen subjects consisted of 8 women and 5 men with an age range of 16 years to 17 years. All subjects were known to have no sleep disorders and psychological disorders. And also they are prohibited from consuming food or drinks that contain caffeine and drugs in the 24 hours before measurement.

#### *B. Experimental Design*

The stimulus used in this study consisted of 3 types of cognitive tasks, namely the arithmetic task, the task of summarizing an article, and the maze problem. For each task, use a black background and white for letters and numbers. For arithmetic assignments, a time of 300 seconds (5 minutes) is provided, using the addition or subtraction or multiplication or division operator (+; -; ×; ÷) of two variables. Each question that appears is only given 8 seconds to answer. If the question is not answered immediately, then the question will change. And for the task of summarizing an article, 10 minutes are available consisting of 3 minutes reading the article and 7 minutes summarizing the article. Finally, the maze problem consists of 3 types of maze with different levels of difficulty, each maze is provided with a completion time of 85 seconds. Thus, this stimulus takes a total time of 20 minutes.

#### *C. Experimental Procedure*

In this measurement, we use 2 computer devices, one computer placed in front of the subject with a distance of approximately 40 centimeters. The stimulus (cognitive task) is displayed on the computer monitor screen. Whereas the EEG device mounted on the subject's head is connected via Bluetooth with another computer, which is used to record the subject's brain activity. It is shown in Fig.1. Before the subject starts working on the given stimulus, subjects will be asked to fill in the IFS questionnaire, this results of the IFS questionnaire will be used as data questionnaire before measurement.

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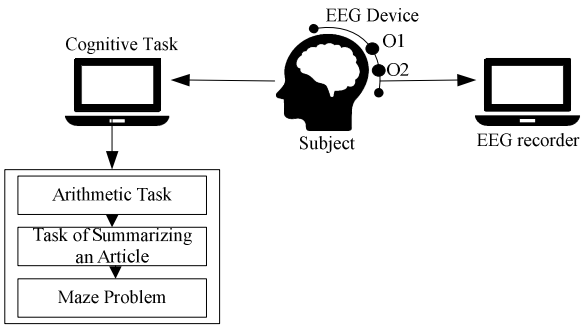


Fig. 1. Measurement Scheme

The subject will use the EEG EMOTIV EPOC+ device. Afterward, the EEG will record the brain waves of the subjects in their closed eyes for 3 minutes, which we call pre-task data. Next, the subject will work on the stimulus. After it finished, subjects were asked to close their eyes once more to record the last brainwave for 3 minutes, which we call data post-task. Finally, subjects were asked to fill out the IFS questionnaire once again; this results of the IFS questionnaire were used as data questionnaires after measurement. To maintaining the subject's concentration, they were asked to focus their attention on the 3 types of cognitive tasks given during the measurement process.

The measurements were taken two times. The first carried out in the morning while the subject was assumed to be in fresh condition. The second measurement was carried out in the afternoon after school activities ended with speculation that the subject was in a state of mental fatigue. The cognitive tasks used in this study are new tasks for all subjects.

### D. Signal Denoising

From the raw EEG data obtained, each subject for one EEG signal measurement consists of 2 kinds of data, namely pre-task data and post-task data. One type of data consists of 2 channel locations for 3 minutes of measurements (eyes closed), and with a sampling frequency of 256 Hz. So one type of data produces 92.160 data lengths and if 2 types of data for subject measurement it will produce 184.320 data lengths. Because the measurement for each subject is done twice, it will produce 4 kinds of measurements with 368.640 data length. For a summary, it can be seen in Table I.

TABLE I. DESCRIPTION OF DATA MEASUREMENT FOR EACH SUBJECT

| | Type of Measurement | Data Size for Each Subject | Description |
|---|---|---|---|
| 1th | Pre-task data | 2 × 3 × 60 × 256 | Number of channel × 3 minutes for eyes close × frequency sampling (Hz) |
| | Post-task data | 2 × 3 × 60 × 256 | Number of channel × 3 minutes for eyes close × frequency sampling (Hz) |
| 2nd | Pre-task data | 2 × 3 × 60 × 256 | Number of channel × 3 minutes for eyes close × frequency sampling (Hz) |
| | Post-task data | 2 × 3 × 60 × 256 | Number of channel × 3 minutes for eyes close × frequency sampling (Hz) |
| | Total data | 368.640 | Total data for 1 subject |

Before the feature extraction stage, one pre-task or post-task data in 3 minutes is cut every one minute, with a 256 data frame in 1. The main stages are needed to obtain relevant information from EEG signal recording, namely the preprocessing and feature extraction stages [20]. By using EEGLAB, the preprocessing stage is performed before the feature extractions stage. EEG signals usually still have a variety of artifacts and disorders caused by eye blinks, eye movements, electrical and physiological noise and muscle activity [6][15]. The use of EEGLAB by utilizing the Independent Component Analysis (ICA) technique is considered capable of removing artifacts in the EEG signals [6][14]. Before using the ICA technique, recorded EEG data were filtered using Finite Impulse Response (FIR) bandpass filter (0,2 - 30 Hz) to remove DC offset and limit the desired frequency.

### E. Features Extraction

Extracting relevant features is one of the most important steps for analyzing EEG signal data. Although the time domain features of the EEG signal are not dominant, there are still many approaches to identifying the characteristics of time series that vary between mental fatigue conditions.

As shown in Fig. 2, at this stage, segmentation is performed on recorded EEG signals that have been cleaned of artifacts using the ICA techniques. The EEG signal is segmented into 3 waves (theta, alpha, and beta). The Butterworth filter is used for segmentation. The mean absolute value (MAV) and standard deviation values of amplitude in theta, alpha, and beta waves are used as features extracted from the EEG signal. The values of features were calculated in the first minute of each EEG signal data.

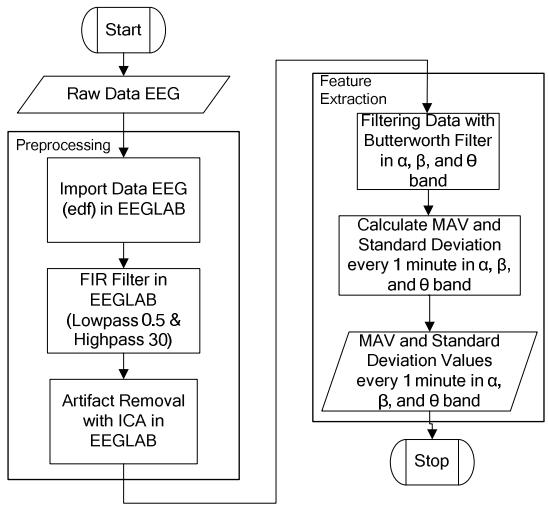


Fig. 2. EEG Signal Processing Flowchart

This *n-th* order Butterworth filter is used in the EEG data segmentation process. Because this filter has a flat response in the required frequency range and constant shape increase *n* [15]. The general formula for Butterworth filters depends on whether the order is odd or even. For an even order, the formula is:

$$T(s) = \prod_{1}^{M/2}\left(\frac{{\omega_N}^2}{s^2 + 2\cos(\theta_i)\omega_N s + {\omega_N}^2}\right), \theta_i = (i - 0.5) \times 180/N \quad (1)$$

The symbol Π indicates a series of products, similar to the way Σ indicates a sum. M indicates filter order. For $\omega_N$ indicates minimum filter bandwidth.

In statistics, the standard deviation (σ) is a measure of the amount of variation or dispersion of a set of values. A low standard deviation indicates that the value tends to be close to the mean of the set, while a high standard deviation indicates that the values are spread out over a wider range. The formula can be expressed as:

$$\sigma_X = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(X(t) - \mu_X)^2} \quad (2)$$

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The MAV feature, specifically, the sum of the values divided by the absolute number of values. The formula can be expressed as:

$$\mu_T = \left|\frac{1}{T}\sum_{t=1}^{T}\mu(t)\right| \quad (3)$$

From these features, the value of each feature is obtained in the first minute, both pre-task and post-task data and three waves (theta, alpha, and beta). The ANOVA analysis with a significance level of 0.05 was used as a significant test of the difference between normal and mental fatigue conditions.

## III. Result and Discussion

The results of the IFS questionnaire on the second measurement showed that from 13 measured subjects turned out to be only 5 subjects suffered from mental fatigue. While 8 subjects did not experience mental fatigue during the second measurement with an IFS questionnaire value of less than 30. This is because 8 subjects may not easily experience mental fatigue. Also, the cognitive tasks given are also considered too easy for them to make them experience mental fatigue. Therefore, the data analysis and discussion only included data from the 5 subjects.

TABLE II. IOWA Fatigue Scale (IFS) Questionnaire Result

| Subject | First Measurement | | Second Measurement | |
|---|---|---|---|---|
| | Before | After | Before | After |
| 1 | 26 | 28 | 38 | 37 |
| 2 | 23 | 23 | 31 | 31 |
| 3 | 18 | 22 | 16 | 37 |
| 4 | 29 | 35 | 30 | 33 |
| 5 | 23 | 25 | 35 | 37 |

Table II shows the results of the IFS questionnaire on 5 subjects. According to IFS, subjects experienced from mental fatigue when the value of the IFS questionnaire was at values 30 to 39. From table II, it can be seen that subject 3 experienced mental fatigue after the cognitive tasks were given in the second measurement with an IFS value of 37. Subject 4 tended to experience mental fatigue because subject 4 experienced mental fatigue after the cognitive tasks were given in the first measurement with an IFS value of 35 and the second with an IFS value of 33.

TABLE III. MAV Feature in Beta Wave on O1

| Subject | Normal | | Difference Post & Pre | Mental Fatigue | | Difference Post & Pre |
|---|---|---|---|---|---|---|
| | Pre-Task | Post-Task | | Pre-Task | Post-Task | |
| 1 | 2,37 | 3,25 | 0,88 | 1,2 | 3,08 | 1,88 |
| 2 | 3,88 | 4,17 | 0,29 | 4,13 | 32,15 | 28,02 |
| 3 | 4,36 | 4,98 | 0,62 | 3,72 | 5,94 | 2,22 |
| 4 | 2,62 | 1,31 | -1,31 | 4,24 | 7,59 | 3,35 |
| 5 | 1,94 | 4,33 | 2,39 | 4,95 | 5,69 | 0,74 |

From Table III, information about beta wave activity with MAV features was obtained in 5 subjects on the O1 channel. When compared with normal conditions, for the MAV feature values between pre-task and post-task there was a higher increase when subjects 1, 2, 3 and 4 experienced mental fatigue. In subject 5 there was also an increase in the value of the MAV feature between pre-task and post-task data, but the increase was greater in normal conditions. But the post-task and pre-task values in mental fatigue are still higher than normal conditions. This increase can be seen from the column difference between post-task and pre-task.

TABLE IV. MAV Feature in Beta Wave on O2

| Subject | Normal | | Difference Post & Pre | Mental Fatigue | | Difference Post & Pre |
|---|---|---|---|---|---|---|
| | Pre-Task | Post-Task | | Pre-Task | Post-Task | |
| 1 | 3,8 | 4,49 | 0,69 | 3,32 | 4,91 | 1.59 |
| 2 | 7,68 | 7,75 | 0,07 | 6,82 | 7,46 | 0.64 |
| 3 | 5,5 | 6,75 | 1,25 | 7,87 | 7,27 | -0.6 |
| 4 | 8,87 | 5,72 | -3,15 | 13,34 | 12,63 | -0.71 |
| 5 | 5,22 | 6,23 | 1,01 | 4,3 | 3,72 | -0.58 |

Table IV provides information on beta wave activity with the MAV feature on the O2 channel. When experiencing mental fatigue, only subjects 1 and 2 experienced a high increase between pre-task and post-task with values of 1.59 and 0.64. Under normal conditions, the MAV feature values in subjects 1 and 2 only increased by 0.69 and 0.07. Different things occur in subjects 3, 4, and 5 when experiencing mental fatigue, where there is a decrease in the value of the MAV feature between post-task and pre-task. Whereas during normal conditions, they experience an increase in the value of the MAV feature between post-task and pre-task. Despite the decline, the value of the MAV feature during post-task and pre-task remains higher than normal conditions.

TABLE V. Standard deviation Feature in Beta Wave on O1

| Subject | Normal | | Difference Post & Pre | Mental Fatigue | | Difference Post & Pre |
|---|---|---|---|---|---|---|
| | Pre-Task | Post-Task | | Pre-Task | Post-Task | |
| 1 | 3,32 | 4,38 | 1,06 | 1,8 | 4,16 | 2,36 |
| 2 | 5,07 | 5,76 | 0,69 | 5,6 | 45,63 | 40,03 |
| 3 | 5,79 | 6,76 | 0,97 | 5,16 | 7,95 | 2,79 |
| 4 | 3,54 | 1,76 | -1,78 | 5,68 | 11,54 | 5,86 |
| 5 | 2,61 | 5,79 | 3,18 | 6,57 | 7,7 | 1,13 |

Table V shows the results of the calculation of the standard deviation features on the beta wave for the O1 channel. When subjects 1, 2, 3, and 4 showed a state of mental fatigue, the value of the standard deviation between the post-task and pre-task increased significantly when compared to normal conditions. In subject 5, the standard deviation feature between the post-task and pre-task increased lower than normal conditions. But the value of pre-task (6,57) and post-task (7,7) in the state of mental fatigue is still higher than normal conditions.

TABLE VI. Standard deviation Feature in Beta Wave on O2

| Subject | Normal | | Difference Post & Pre | Mental Fatigue | | Difference Post & Pre |
|---|---|---|---|---|---|---|
| | Pre-Task | Post-Task | | Pre-Task | Post-Task | |
| 1 | 5,24 | 5,97 | 0,73 | 4,84 | 6,91 | 2,07 |
| 2 | 9,99 | 10,7 | 0,71 | 9,3 | 10,1 | 0,8 |
| 3 | 7,41 | 9,15 | 1,74 | 10,94 | 9,73 | -1,21 |
| 4 | 11,93 | 8,84 | -3,09 | 18,27 | 17,08 | -1,19 |
| 5 | 7,01 | 8,15 | 1,14 | 5,7 | 5,02 | -0,68 |

The results of the calculation of standard deviation features in the beta wave for the O2 channel are shown in Table VI. Subjects 1 and 2 experienced a high increase between post-task and pre-task by 2,07 and 0,8 during mental fatigue. The lower increase in subjects 1 and 2 occur during normal conditions with a different value of 0,73 and 0,71 between post-task and pre-task. In subjects 3 and 4 there was a decrease in the standard deviation feature when mental fatigue between post-task and pre-task with values -1,21 and -1,19. But when experiencing mental fatigue, the value of pre-task and post-task on subjects 3 and 4 is still higher compared to normal conditions. Meanwhile, subject 5 experienced a decrease in the standard deviation feature between post-task and pre-task during mental fatigue. Even the pre-task and post-task scores on subject 5 when mental fatigue is lower than normal conditions.

TABLE VII. MAV Feature in Alpha Wave on O1

| Subject | Normal | | Difference Post & Pre | Mental Fatigue | | Difference Post & Pre |
|---|---|---|---|---|---|---|
| | Pre-Task | Post-Task | | Pre-Task | Post-Task | |
| 1 | 2,17 | 2,38 | 0,21 | 0,83 | 2,05 | 1,22 |
| 2 | 6,8 | 7,6 | 0,8 | 5,02 | 24,37 | 19,35 |
| 3 | 4,53 | 5,12 | 0,59 | 3,12 | 5,6 | 2,48 |
| 4 | 3,03 | 0,89 | -2,14 | 3,95 | 9,53 | 5,58 |
| 5 | 1,02 | 2,57 | 1,55 | 2,62 | 3,54 | 0,92 |

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The results of calculating MAV features on alpha waves for the O1 channel are shown in Table VII. When compared with normal conditions, subjects 1, 2, 3, and 4 experienced a high increase in the MAV feature when suffering from mental fatigue between post-task and pre-task. Subject 5 also experienced an increase in the value of the MAV feature between post-task and pre-task when suffering from mental fatigue with a value of 0,92. Although the increase in the value of the MAV feature is lower than the increase in normal conditions with a value of 1,55. But the post-task value in mental fatigue conditions is higher than the post-task value in normal conditions.

TABLE VIII. MAV FEATURE IN ALPHA WAVE ON O2

| Subject | Normal | | Difference | Mental Fatigue | | Difference |
|---|---|---|---|---|---|---|
| | Pre-Task | Post-Task | Post & Pre | Pre-Task | Post-Task | Post & Pre |
| 1 | 4,33 | 4,45 | 0,12 | 2,29 | 4,42 | 2,13 |
| 2 | 14,06 | 13,48 | -0,58 | 9,3 | 10,54 | 1,24 |
| 3 | 6,43 | 7,2 | 0,77 | 6,46 | 7,27 | 0,81 |
| 4 | 10,17 | 4,99 | -5,18 | 11,83 | 12,08 | 0,25 |
| 5 | 2,9 | 3,65 | 0,75 | 2,99 | 2,29 | -0,7 |

The results of the MAV feature calculation on the alpha wave for the O2 channel are shown in Table VIII. Subjects 1, 2, 3, and 4 when suffering from mental fatigue experienced a high increase in MAV features between post-task and pre-task when compared with normal conditions. In contrast, subject 5 experienced a decrease in the MAV feature between post-task and pre-task when suffering from mental fatigue with a value of -0,7. Even the pre-task and post-task values are lower than normal conditions with values of 2,99 and 2,29. An increase in the value of features MAV on subject 5 only occurs during normal conditions.

TABLE IX. STANDARD DEVIATION FEATURE IN ALPHA WAVE ON O1

| Subject | Normal | | Difference | Mental Fatigue | | Difference |
|---|---|---|---|---|---|---|
| | Pre-Task | Post-Task | Post & Pre | Pre-Task | Post-Task | Post & Pre |
| 1 | 3,34 | 3,57 | 0,23 | 1,35 | 3,38 | 2,03 |
| 2 | 10,86 | 12,52 | 1,66 | 8,01 | 39,92 | 31,91 |
| 3 | 7,25 | 8,16 | 0,91 | 5,01 | 8,97 | 3,96 |
| 4 | 4,8 | 1,39 | -3,41 | 6,07 | 15,52 | 9,45 |
| 5 | 1,63 | 3,68 | 2,05 | 4,16 | 5,64 | 1,48 |

Table IX shows the results of the calculation of the standard deviation features on alpha waves for the O1 channel. From this table, it can be seen that subjects 1, 2, 3, and 4 experienced a high increase in the standard deviation between post-task and pre-task when suffering from mental fatigue compared to the increase in normal conditions. Subject 5 also experienced an increase in the standard deviation feature between post-task and pre-task when suffering from mental fatigue by a value of 1,48. But the increase in the standard deviation was lower than the increase that occurred during normal conditions with a value of 2,05.

TABLE X. STANDARD DEVIATION FEATURE IN ALPHA WAVE ON O2

| Subject | Normal | | Difference | Mental Fatigue | | Difference |
|---|---|---|---|---|---|---|
| | Pre-Task | Post-Task | Post & Pre | Pre-Task | Post-Task | Post & Pre |
| 1 | 6,84 | 6,87 | 0,03 | 3,75 | 6,81 | 3,06 |
| 2 | 22,08 | 22,54 | 0,46 | 15,05 | 17,39 | 2,34 |
| 3 | 10,13 | 11,76 | 1,63 | 10,43 | 11,59 | 1,16 |
| 4 | 16,29 | 9,46 | -6,83 | 18,43 | 18,53 | 0,1 |
| 5 | 4,75 | 5,39 | 0,64 | 4,72 | 3,62 | -1,1 |

For the O2 channel, the results of the calculation of standard deviation features on alpha waves are shown in Table X. Subjects 1, 2, and 4 experienced a high increase in the standard deviation features between post-task and pre-task when the IFS questionnaire showed they suffered from mental fatigue. For subjects 1, 2, and 4, the increase in the value of the standard deviation between the post-task and pre-task is even higher than normal conditions. Meanwhile, in subject 3, the standard deviation feature value also increased between post-task and pre-task when suffering from mental fatigue by a value of 1,16. But the increase of 1,16 was lower than normal conditions with a value of 1,63. On the contrary, when suffering from fatigue, mental subject 5 has decreased the standard deviation feature between post-task and pre-task. Even the value of pre-task and post-task is lower than normal conditions.

TABLE XI. MAV FEATURE IN THETA WAVE ON O1

| Subject | Normal | | Difference | Mental Fatigue | | Difference |
|---|---|---|---|---|---|---|
| | Pre-Task | Post-Task | Post & Pre | Pre-Task | Post-Task | Post & Pre |
| 1 | 1,9 | 1,81 | -0,09 | 1,08 | 2,61 | 1,53 |
| 2 | 3,29 | 3,52 | 0,23 | 2,87 | 31,56 | 28,69 |
| 3 | 1,79 | 1,86 | 0,07 | 1,55 | 3 | 1,45 |
| 4 | 1,47 | 1,01 | -0,46 | 1,7 | 4,32 | 2,62 |
| 5 | 0,82 | 2,81 | 1,99 | 1,91 | 2,14 | 0,23 |

The results of the MAV feature on the theta wave for the O1 channel are shown in Table XI. Compared to normal conditions, Subjects 1, 2, 3, and 4 experienced a high increase in MAV feature values between post-task and pre-task when the IFS questionnaire showed subjects suffering from mental fatigue. Meanwhile, subject 5 also experienced an increase in the value of the MAV feature between post-task and pre-task when suffering from mental fatigue by a value of 0,23. But the increase in the value of the MAV feature under conditions of mental fatigue was lower than normal conditions. A high increase of 1,99 for the MAV feature value between the post-task and pre-task under normal conditions.

TABLE XII. MAV FEATURE IN THETA WAVE ON O2

| Subject | Normal | | Difference | Mental Fatigue | | Difference |
|---|---|---|---|---|---|---|
| | Pre-Task | Post-Task | Post & Pre | Pre-Task | Post-Task | Post & Pre |
| 1 | 2,28 | 2,34 | 0,06 | 2,48 | 5,08 | 2,6 |
| 2 | 6,9 | 7,27 | 0,37 | 6,57 | 7,65 | 1,08 |
| 3 | 1,9 | 2,43 | 0,53 | 3,07 | 3,28 | 0,21 |
| 4 | 3,28 | 3,08 | -0,2 | 4,93 | 5,52 | 0,59 |
| 5 | 1,99 | 3,78 | 1,79 | 1,39 | 1,3 | -0,09 |

For the O2 channel, the results of the MAV feature calculation on the theta wave are shown in Table XII. When compared with the increase that occurred between post-task and pre-task under normal conditions, Subjects 1, 2, 3, and 4 experienced a high increase in the value of the MAV feature between post-task and pre-task in a state of mental fatigue. On the other hand, subject 5 experienced a decrease in the MAV feature between post-task and pre-task when the subject's IFS questionnaire showed mental fatigue. The pre-task and post-task scores of subject 5 were also lower when suffering from mental fatigue compared to normal conditions.

TABLE XIII. STANDARD DEVIATION FEATURE IN THETA WAVE ON O1

| Subject | Normal | | Difference | Mental Fatigue | | Difference |
|---|---|---|---|---|---|---|
| | Pre-Task | Post-Task | Post & Pre | Pre-Task | Post-Task | Post & Pre |
| 1 | 2,87 | 2,75 | -0,12 | 1,72 | 4,52 | 2,8 |
| 2 | 4,94 | 5,34 | 0,4 | 4,28 | 53,76 | 49,48 |
| 3 | 2,62 | 2,76 | 0,14 | 2,27 | 4,43 | 2,16 |
| 4 | 2,23 | 1,53 | -0,7 | 2,54 | 11,56 | 9,02 |
| 5 | 1,26 | 3,8 | 2,54 | 2,87 | 3,21 | 0,34 |

Table XIII shows the calculation results of the standard deviation features on the theta wave for the O1 channel. From

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table XIII it can be seen that there is an increase in features of high standard deviation between the post-task and pre-task values in subjects 1, 2, 3, and 4 when they experience mental fatigue. The increase in features between the post-task and pre-task is lower than when the subjects were in normal conditions. Meanwhile, subject 5 also experienced an increase between post-task and pre-task when experiencing mental fatigue by a value of 0,34. But the value is lower than the increase that occurred between post-task and pre-task in normal conditions by a value of 2,54.

TABLE XIV. STANDARD DEVIATION FEATURE IN THETA WAVE ON O2

| Subject | Normal | | Difference Post & Pre | Mental Fatigue | | Difference Post & Pre |
|---|---|---|---|---|---|---|
| | Pre-Task | Post-Task | | Pre-Task | Post-Task | |
| 1 | 3,42 | 3,43 | 0,01 | 3,98 | 7,99 | 4,01 |
| 2 | 11,49 | 12,5 | 1,01 | 8,18 | 11,17 | 2,99 |
| 3 | 2,64 | 3,38 | 0,74 | 4,63 | 4,72 | 0,09 |
| 4 | 4,69 | 4,58 | -0,11 | 7,48 | 8,8 | 1,32 |
| 5 | 3,07 | 5,27 | 2,2 | 2,1 | 1,98 | -0,12 |

For the O2 channel, the calculation results of the standard deviation features on the theta wave are shown in Table XIV. From this table, it can be seen that subjects 1, 2, 3, and 4 experienced a high increase in the value of standard deviation features between post-task and pre-task when all four subjects experienced mental fatigue. The lower increase between post-task and pre-task occurs when all four subjects are in normal condition. Subject 5 experienced a decrease in the standard deviation feature between post-task and pre-task when subject 5 experienced mental fatigue. An increase in the standard deviation feature values between the post-task and pre-task occurs when subject 5 is in normal condition.

In general, the four subjects experienced an increase in the value of MAV features and standard deviations in the beta, alpha, and theta waves between post-task and pre-task when suffering from mental fatigue. This shows the activity of the three waves increases when the subject suffers from mental fatigue. This proves that the subject is trying hard to maintain a level of alertness [4][16], drowsiness, and decreased cognitive performance [4]. Decreased cognitive performance is indicated by the results of the subject's cognitive tasks, for example, where subject 3 is in normal condition able to answer 140 arithmetic questions, but subject 3 is only able to answer 122 arithmetic questions when experiencing mental fatigue. In the O2 channel, subject 5 tended to experience a decrease in theta, alpha, and beta waves activity when suffering from mental fatigue. This is possible because the right brain activity (O2) is not too high compared to the left (O1).

TABLE XV. STATISTICAL RESULT OF MAV FEATURE ON 5 SUBJECTS

| Wave | Channel O1 | | | Channel O2 | | |
|---|---|---|---|---|---|---|
| | Correlation | $R^2$ | *p*-value | Correlation | $R^2$ | *p*-value |
| Beta | 0,477 | 0,227 | 0,416 | 0,853 | 0,727 | 0,066 |
| Alpha | 0,862 | 0,742 | 0,060 | 0,893 | 0,797 | 0,042 |
| Theta | 0,892 | 0,796 | 0,042 | 0,825 | 0,680 | 0,086 |

The significant was measured with *p*-value < 0,05

Table XV illustrates the statistical results of the MAV feature of beta, alpha, and theta waves. The results of measurements on the O1 channel in theta waves explain a lot of variations within the data and there are statistically significant differences between mental fatigue and normal condition with *p*-value<0,05. The measurement results on the O2 channel in alpha waves between mental fatigue and normal condition also describes the large variations in the data and statistically significant differences with *p*-value<0,05. The O1 channel shows a high correlation value between when experiencing mental fatigue and normal conditions in alpha and theta waves. The beta, alpha, and theta waves in the O2 channel also revealed a high correlation value between the two conditions.

TABLE XVI. STATISTICAL RESULT OF STANDARD DEVIATION FEATURE ON 5 SUBJECTS

| Wave | Channel O1 | | | Channel O2 | | |
|---|---|---|---|---|---|---|
| | Correlation | $R^2$ | *p*-value | Correlation | $R^2$ | *p*-value |
| Beta | 0,434 | 0,188 | 0,465 | 0,867 | 0,752 | 0,057 |
| Alpha | 0,861 | 0,742 | 0,061 | 0,924 | 0,855 | 0,025 |
| Theta | 0,890 | 0,792 | 0,043 | 0,756 | 0,571 | 0,140 |

The significant was measured with *p*-value < 0,05

The statistical results of the standard deviation features in beta, alpha, and theta waves between mental fatigue and normal conditions are illustrated by Table XVI. It is concluded that the theta waves can explain a lot of variations in data and significant differences in statistical values (*p*-value < 0,05) by the O1 channel. In the O2 channel, alpha waves show a high variation also a significant difference between the data with a *p*-value < 0,05. However, in O1 Channel, beta waves do not show a lot of variation in data and with large *p*-values (*p*-value = 0,434). Strong data correlations are shown in alpha and theta waves by the O1 channel. Strong data correlations are also presented in beta and alpha waves by the O2 channel.

## IV. CONCLUSION

Based on the results of measurements with EEG and observations with the IFS from 4 subjects in the O1 and O2 occipital areas, showed that subjects who are high school students could suffer mental fatigue after undergoing school activities about 8 hours a day. By using EEG, it was concluded that there was a high increase between the post-task and pre-task of the MAV and the standard deviation features in the theta, alpha, and beta waves when subjects experienced mental fatigue. The lower increase between post-task and pre-task occurs when subjects are in normal condition. When suffering from mental fatigue, an increase in these three waves generally occurs in the occipital areas of O1 and O2. In this study, the use of cognitive tasks can increase the subject's cognitive activity so that subjects easily experience mental fatigue. For further research, the 8 subjects who did not suffer mental fatigue on the second measurement, they could experience mental fatigue if the level of cognitive tasks was higher than the previous one. Because we suspect this cognitive task is easy for the 8 subjects. For the development of this study, we suggest comparing the results of the EEG signal features based on the order of the tasks given. Moreover, to obtain more accurate results, it is best to do several experiments for each subject.

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